

PLANTILLA EXCEL: CÁLCULO IGM Y ROI INDICADORES CIBERSEGURIDAD-IA

Implementación Step-by-Step en Excel

Versión: 1.0 | **Fecha:** Enero 2026

Contexto: Cálculo Índice Gestión Métricas (IGM) + ROI para 9 Indicadores

Software: Microsoft Excel / Google Sheets compatible

INTRODUCCIÓN: IGM (ÍNDICE DE GESTIÓN DE MÉTRICAS)

El IGM es una métrica compuesta (0-100) que integra:

1. **Puntuación PRAGMATIC** (9 criterios, avg)
2. **Madurez GQM** (Nivel 0-5 capturado)
3. **Compliance Normativo** (% cumplimiento vs regulaciones)
4. **Implementación Operacional** (% deployed vs target)

Fórmula IGM:

$$\text{IGM} = \frac{(\text{PRAGMATIC_Score} \times 0.40) + (\text{GQM_Maturity} \times 20) + (\text{Compliance_}\% \times 0.30) + (\text{Implementation_}\% \times 0.10)}{0.40 + 0.30 + 0.10 + 0.20} \times 100$$

Rango Interpretación:

- 0-25: Inicial (Nivel 1-2)
- 26-50: Definido (Nivel 2-3)
- 51-75: Gestionado (Nivel 3-4)
- 76-100: Optimizado (Nivel 4-5)

SECCIÓN 1: ESTRUCTURA PLANTILLA EXCEL

HOJA 1: "Dashboard"

Propósito: Visión ejecutiva consolidada

Estructura Filas/Columnas:

DASHBOARD EJECUTIVO - INDICADORES CIBERSEGURIDAD-IA ESPAÑA 2026

Fecha Actualización: [Celda: A2] = TODAY()
Período: [Q1 2026] Organización: [Editable]

Fila 5-6: Encabezados

A	B	C	D	E	F	G	H	I	J
Indicador	GQM Level (0-5)	PRAGMATIC Avg	Compliance %	Deployment %	IGM Score (0-100)	ROI Year 1 (€)	Payback Period (meses)	Estado	Prioridad

Filas 7-15: Datos por indicador

1. AI-Detection	[3]	[4.3]	[80%]	[60%]	[IGM_FORMULA]	[ROI_FORMULA]	[PAYBACK_FORMULA]	● On Track	P1
2. Vuln.Mgmt	[3]	[4.4]	[70%]	[50%]	[IGM_FORMULA]	[ROI_FORMULA]	[PAYBACK_FORMULA]	● On Track	P1
3. Deepfake	[2]	[3.1]	[40%]	[20%]	[IGM_FORMULA]	[ROI_FORMULA]	[PAYBACK_FORMULA]	● Monitor	P3
4. SIEM+Hunt	[3]	[4.4]	[60%]	[40%]	[IGM_FORMULA]	[ROI_FORMULA]	[PAYBACK_FORMULA]	● On Track	P1
5. Training	[2]	[4.2]	[80%]	[70%]	[IGM_FORMULA]	[ROI_FORMULA]	[PAYBACK_FORMULA]	● On Track	P1
6. Compliance	[2]	[3.7]	[100%]	[80%]	[IGM_FORMULA]	[ROI_FORMULA]	[PAYBACK_FORMULA]	● On Track	P2
7. Supply Chain	[1]	[4.0]	[50%]	[30%]	[IGM_FORMULA]	[ROI_FORMULA]	[PAYBACK_FORMULA]	● Monitor	P3
8. Incident Response	[2]	[4.6]	[60%]	[40%]	[IGM_FORMULA]	[ROI_FORMULA]	[PAYBACK_FORMULA]	● On Track	P1
9. Governance	[3]	[4.2]	[80%]	[90%]	[IGM_FORMULA]	[ROI_FORMULA]	[PAYBACK_FORMULA]	● On Track	P1
PROMEDIO TOTAL	[=AVERAGE(B7:B15)]	[=AVERAGE(C7:C15)]	[=AVERAGE(D7:D15)]	[=AVERAGE(E7:E15)]	[=AVERAGE(F7:F15)]	[=SUM(G7:G15)]	[=AVERAGE(H7:H15)]		

Fila 17: Métricas de Impacto

Métrica	Valor
Portfolio IGM Promedio	[=AVERAGE(F7:F15)]
Investment Total Year 1	€1.9M [=SUM("Costs"!B:B)]
Expected ROI Year 1	26% [(SUM(G7:G15)/1900000)]
Break-even (meses)	18
Cumulative ROI 3-year	€3.2M
Regulatory Compliance Avg	[=AVERAGE(D7:D15)]%

HOJA 2: "Detalle Indicadores"

Propósito: Nivel de detalle por indicador (9 hojas separadas, una por indicador)

Ejemplo para Indicador 1 (AI-Detection):

INDICADOR 1: AI-THREAT DETECTION COVERAGE

FILA 1-4: Metadatos

Nombre Indicador	AI-Threat Detection Coverage
Código	IND-001
GQM Goal	Reducir MTTD 181 días → <24 horas
Status	Green (●)

FILA 6-10: PRAGMATIC Scores (Importado de Matriz PRAGMATIC)

Criterio	Score (1-5)	Evidencia	Fuente
P - Predictivo	5	MTTD reduction = breach cost reduction	CNCS data
R - Relevante	5	NIS2 mandated	Regulatory
A - Accionable	4	Clear thresholds: <24h = goal met	Internal SLA
G - Genuino	5	Mide outcome real (no proxy)	SIEM telemetry
M - Significativo	5	Board-level importance	Strategic plan
P - Preciso	4	Timestamps en SIEM	Technical spec
T - Oportuno	5	Real-time SIEM data	Continuous
I - Independiente	4	Affected by threat sophistication	Threat landscape
C - Rentable	3	SIEM cost €180K annually	Budget
PRAGMATIC AVG	4.3		

FILA 12-16: GQM Madurez (Escala 0-5)

Level 0 - Inexistente	[] Not implemented
Level 1 - Inicial	[] Ad-hoc
Level 2 - Repetible	[] Inconsistent
Level 3 - Definido	[✓] Selected (Current)
Level 4 - Gestionado	[] Cuantitativamente medido
Level 5 - Optimizado	[] Continuous innovation

CURRENT LEVEL: 3

TARGET LEVEL: 5

TIMELINE TO TARGET: 12 meses

FILA 18-22: Compliance Normativo

Normativa	Requisito	Cumplimiento	Gap	Plazo
NIS2 Art.17	Coverage ≥80% essential entities	80% ✓	0%	Agosto 2026
Plan Nacional L4.3	MTTD <24 horas	15% (MTTD 50d actual)	-85%	Q4 2026
ISO 27001 A.16.1.5	Monitoring systems required	70%	-30%	Ongoing
EU AI Act Art.37	Risk assessment documented	60%	-40%	2026
COMPLIANCE AVG		56%		

FILA 24-28: Implementación Operacional (Current State)

Componente	Target	Current	% Deployed	Inversión
SIEM Deployment	100%	70%	70%	€150K invested
ML Correlation Engine	100%	40%	40%	€80K needed
Threat Intel Integration	100%	30%	30%	€50K needed
Alert Tuning/Optimization	100%	20%	20%	€30K needed
DEPLOYMENT AVG			40%	€310K total Phase 1

FILA 30-32: IGM Calculation

Formula Component	Value	Weight	Weighted
PRAGMATIC Score (4.3)	4.3	40%	1.72
GQM Level (3 → 20 pts)	20	30%	6.0
Compliance % (56%)	56	20%	11.2
Deployment % (40%)	40	10%	4.0
IGM SCORE	62.92		= Nivel 4 (Gestionado)

FILA 34-36: Métricas Clave

Métrica	Baseline	Target 2026	Progreso
MTTD (días)	50	<24	En progreso
Coverage %	60%	95%	67% complete
False Positive Rate	30%	<15%	En progreso

HOJA 3: "Costs & ROI"

Propósito: Desglose costos CapEx/OpEx + cálculo ROI

COSTS & ROI ANALYSIS - YEAR 1 (2026)

FILA 1-4: Resumen Ejecutivo ROI

Métrica	Valor
Total Investment Year 1	€1,900,000
Expected Cost Avoidance	€500,000 - €2,000,000
ROI Estimated	26% - 105%
Payback Period	12-18 meses

FILA 6-30: Desglose por Indicador (Capex + Opex)

Indicador	Elemento	Tipo	Cantidad	Unit Cost	Total Year 1
1. AI-Detection					
	SIEM Enhancement	CapEx	1	€150,000	€150,000
	SIEM Annual License	OpEx	1	€100,000	€100,000
	ML Model Training	CapEx	200 hrs	€300/hr	€60,000
	SOC Analyst (0.5 FTE)	OpEx	0.5	€75,000	€37,500
	Subtotal IND-1			**€347,500**	
2. Vuln.Mgmt					
	Scanning Tools (license upgrade)	CapEx	1	€80,000	€80,000
	Annual Maintenance	OpEx	1	€40,000	€40,000
	SBOM Integration	CapEx	1	€60,000	€60,000
	VA Analyst (0.5 FTE)	OpEx	0.5	€75,000	€37,500
	Subtotal IND-2			**€217,500**	
3. Deepfake					
	Pilot Technology (first year)	CapEx	1	€200,000	€200,000
	Deployment (deferred to 2027)	CapEx	0	-	€0
	Subtotal IND-3			**€200,000**	
4. SIEM+Hunting					
	Threat Hunting Platform	CapEx	1	€150,000	€150,000
	Annual License	OpEx	1	€80,000	€80,000
	Threat Hunting Team (1.5 FTE)	OpEx	1.5	€75,000	€112,500
	Subtotal IND-4			**€342,500**	
5. Training					
	Phishing Simulator Upgrade	CapEx	1	€40,000	€40,000
	Annual License	OpEx	1	€25,000	€25,000
	Training Content Development	CapEx	1	€15,000	€15,000
	Subtotal IND-5			**€80,000**	
6. Compliance					
	NIS2 Assessment (external)	CapEx	1	€80,000	€80,000
	Compliance Officer (0.3 FTE)	OpEx	0.3	€75,000	€22,500
	EU AI Act Gap Analysis	CapEx	1	€50,000	€50,000
	Subtotal IND-6			**€152,500**	
7. Supply Chain					
	SBOM Tools	CapEx	1	€120,000	€120,000
	Vendor Assessment Program	CapEx	1	€40,000	€40,000
	Subtotal IND-7			**€160,000**	
8. Incident Response					
	SOAR Platform	CapEx	1	€200,000	€200,000
	Annual Subscription	OpEx	1	€100,000	€100,000
	Playbook Development	CapEx	200 hrs	€300/hr	€60,000
	IR Team (0.5 FTE)	OpEx	0.5	€75,000	€37,500
	Subtotal IND-8			**€397,500**	
9. Governance					
	Governance Assessment	CapEx	1	€30,000	€30,000

		Board Reporting Tools		CapEx		1		€15,000		€15,000	
		Subtotal IND-9				**€45,000**					
		TOTAL CAPEX (2026)				**€1,155,000**					
		TOTAL OPEX (2026)				**€745,000**					
		TOTAL INVESTMENT (2026)				**€1,900,000**					

FILA 32-50: ROI Calculation

	Scenario		Baseline Assumption		Calculation	
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	CONSERVATIVE Scenario		1 breach prevented (annual)			
	Incident Cost Avoided		1 × €500,000 =		€500,000	
	Implementation Cost				€1,900,000	
	Net Benefit Year 1				-€1,400,000	
	Year 2-3 Cumulative		€500K × 2 additional breaches prevented =		€1,000,000	
	3-Year ROI				-€400,000 (break-even year 4)	
	REALISTIC Scenario		40% incident reduction + 20% faster response			
	Incident Cost Avoidance		2 incidents prevented × €500K =		€1,000,000	
	Response Time Savings		100 incidents × €10K cost reduction =		€1,000,000	
	Total Annual Benefit				€2,000,000	
	Implementation Cost				€1,900,000	
	Net Benefit Year 1				€100,000	
	Year 2-3 Cumulative		€2M × 2 years =		€4,000,000	
	3-Year ROI				110%	
	Payback Period				~11 months	
	OPTIMISTIC Scenario		Industry peer benchmark (Gartner)			
	Incident Cost Avoidance		3 incidents prevented × €750K avg =		€2,250,000	
	Response Efficiency Gains		150 hrs analyst time saved × €300/hr =		€45,000	
	Regulatory Penalty Avoidance		Estimated €250K-€500K range =		€250,000	
	Total Annual Benefit				€2,545,000	
	Net Benefit Year 1				€645,000	
	Payback Period				~9 months	

FILA 52-54: Sensitivity Analysis

	Variable		-20%		Base		+20%		Impact on ROI	
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	Incident Cost		€400K		€500K		€600K		±€200K on Year 1	
	Breach Probability		-40% reduction		-20%		0%		±€1M on Year 1	
	Implementation Timeline		+3 months		On-track		-2 months		±€300K on Year 1	

HOJA 4: "Maturity Trajectory"

Propósito: Proyección de madurez 18-24 meses

MATURITY TRAJECTORY PROJECTION (2026-2027)

FILA 1-4: Timeline Overview

Fase	Período	Indicadores	Target IGM	Investment
Phase 1: Foundation	Q1 2026	9 (Governance baseline)	50	€0 (internal)
Phase 2: Critical Immediate	Q1-Q2 2026	1,2,4,5,8 + 9 cont.	60	€1.2M
Phase 3: Mandated	Q2-Q3 2026	6 (Compliance)	70	€150K
Phase 4: Secondary	Q3-Q4 2026	3 (Deepfake)	65	€500K
Phase 5: Advanced	Q4 2026-Q1 2027	7 (Supply Chain)	75	€350K
Phase 6: Optimization	2027	All 9 integrated	85	€400K

FILA 6-15: Quarter-by-Quarter IGM Projection

Indicador	Q4 2025 (Baseline)	Q1 2026	Q2 2026	Q3 2026	Q4 2026	Q1 2027	Q2 2027
1. AI-Detection	35	45	55	65	75	80	85
2. Vuln.Mgmt	40	50	60	70	75	80	85
3. Deepfake	15	20	25	30	40	50	60
4. SIEM+Hunt	30	40	55	70	75	80	85
5. Training	45	55	65	70	75	80	85
6. Compliance	50	60	75	85	90	92	95
7. Supply Chain	20	25	30	40	55	70	75
8. Incident Response	35	50	65	75	80	85	90
9. Governance	55	70	75	80	85	88	90
PORTFOLIO AVG	**38**	**46**	**57**	**69**	**76**	**81**	**86**

FILA 17-19: Interpretación por Nivel

Portfolio IGM	Nivel	Características
0-25	1 - Inicial	Ad-hoc; no processes; high risk
26-50	2 - Repetible	Processes emerging; inconsistent execution
51-75	3 - Definido	Formal processes; consistent execution; board aware
76-100	4-5 - Gestionado/Optimizado	Automated; metrics-driven; continuous improvement

FILA 21-25: Key Milestones

Hito	Fecha	Indicador(es)	Criterio Éxito	Responsable
Governance Formalization	31-Mar-2026	9	CISO reporting to CEO	CISO
NIS2 Assessment Complete	15-Jun-2026	1,2,6	80% compliance documented	Risk Head
SIEM Operationalization	30-Jun-2026	1,4	<24h MTTD for 80% of incidents	CTO
Phishing Campaign Launch	15-Apr-2026	5	Baseline click-rate <15%	Security Awareness Lead
Incident Response First Playbook	31-May-2026	8	MTTR <1 hour (automated)	IR Lead
EU AI Act Gap Resolution	31-Aug-2026	6	90% gaps remediated	Compliance Officer
Supply Chain SBOM Pilot	30-Sep-2026	7	50% vendor SBOMs collected	Procurement Lead

HOJA 5: "Raw Data Input"

Propósito: Entrada de datos brutos para cálculos

RAW DATA INPUT - MEASUREMENT SHEET

Instrucciones: Actualizar mensualmente con datos más recientes

FILA 1-5: Metadata

Parameter	Value
Organization	[ORGANIZATION NAME]
Assessment Date	[DATE]
Assessor	[NAME]
Review Status	[Draft / Final]

FILA 7-50: Metric Raw Values (por indicador)

Indicador 1: AI-Detection

Métrica	Formula	Current Value	Data Source	Last Update
Coverage % (M1.1)	(Assets con ML-IDS / Total assets críticos) × 100	60%	SIEM inventory	Jan 2026
MTTD días (M1.2)	Avg(Detection_time - Event_time)	50 días	Incident logs	Jan 2026
FP Rate % (M1.3)	(False_alerts / Total_alerts) × 100	28%	SIEM metrics	Jan 2026
Model Accuracy % (M1.4)	Correct_predictions / Total	87%	Model testing	Jan 2026

Indicador 2: Vulnerability Management

Métrica	Formula	Current Value	Data Source	Last Update
Scanning Coverage % (M2.1)	(Systems scanned / Total systems) × 100	55%	VA tools	Jan 2026
MTTR días (M2.2)	Avg(Remediation - Detection)	22 días	Ticketing	Jan 2026
SLA Compliance % (M2.3)	(On-time / Total) × 100	72%	Ticketing	Jan 2026
Risk Score (M2.4)	(Criticality × Exploitability × Threat) / 10	6.5	Risk platform	Jan 2026

[... Repeat para Indicadores 3-9 ...]

SECCIÓN 2: FORMULAS EXCEL PREDEFINIDAS

Fórmula IGM (Copiar en celda F7 para Indicador 1)

=ROUND(((C7*0.4)+(B7*20)+(D7*0.3)+(E7*0.1))*100/1,0)

Donde:

- C7 = PRAGMATIC Score (4.3 en Indicador 1)
- B7 = GQM Level converted to points (Nivel 3 = 20 puntos)

- D7 = Compliance % (80% en Indicador 1)

- E7 = Deployment % (60% en Indicador 1)

Resultado: IGM = 63 (Nivel 3-4: Definido/Gestionado)

Fórmula ROI Year 1

$$=(\text{SUM}(G7:G15) - 1900000) / 1900000$$

Donde:

- G7:G15 = Beneficios por indicador (incident cost avoidance)

- 1,900,000 = Investment total

Resultado: ROI % = $(2,000,000 - 1,900,000) / 1,900,000 = 5.3\%$

Fórmula Payback Period (meses)

$$=\text{ROUND}(12 / ((G7:G15 / 12) / 1900000), 1)$$

Resultado: ~11 meses (REALISTIC scenario)

Fórmula Portfolio IGM Avg

$$=\text{AVERAGE}(F7:F15)$$

SECCIÓN 3: INSTRUCCIONES IMPLEMENTACIÓN

Paso 1: Crear Estructura Base en Excel

1. Descargar template en blanco
2. Crear 5 hojas: Dashboard, Detalle, Costs, Trajectory, Raw Data
3. Copiar encabezados desde este documento

Paso 2: Alimentar Datos Brutos

4. En hoja "Raw Data Input", entrar valores actuales para cada métrica
5. Utilizar timestamps SIEM, ticketing systems, audit reports como source of truth
6. Validar datos con stakeholders (CISO, CTO, Risk Head)

Paso 3: Ejecutar Cálculos

7. Abrir hoja "Dashboard"
8. Fórmulas IGM (F7:F15) calcularán automáticamente
9. ROI (G7:G15) calcularán basados en Costs sheet
10. Payback period (H7:H15) calcularán automáticamente

Paso 4: Actualizar Mensualmente

- Q1 2026: Baseline measurements
- Q2 2026: Re-assessment post Phase 2 implementation
- Q3-Q4 2026: Continuous tracking
- 2027: Advanced optimization tracking

FIN DE PLANTILLA EXCEL